Brian Wonjun Lee

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Education

University of Pennsylvania Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) *June 2026 – Aug 2026 · San Francisco (Remote)*

- **Made the company's AI agent usable by studios outside the company** — packaged it as a typed **SDK** with versioned APIs and **136 tests**, then proved it inside a game mode an **external builder authored**, loading their content at runtime
- **Redirected the team's roadmap** with a **Python multi-agent tool on the Claude Agent SDK** that mined production data through read-only MCP — it showed a reported retention gain was a measurement artifact and surfaced the real churn drivers
- **Let a builder hold a spoken conversation with the in-world character from a browser**, no game client — took hosted cloud speech-to-text and text-to-speech end to end
- **Turned a hand-approved retention channel into an autonomous one** delivering to **91% of targeted devices**, which produced the first revenue the channel ever generated

Software Engineer Intern, BitMango (Puzzle1 Studio) *June 2025 – Aug 2025 · Pangyo, South Korea*

- **Kept 50+ live Unity mobile titles shipping** across a summer of releases, holding UI/UX and gameplay stable
- **Cleared 100+ QA-reported bugs into production**, in Unity prefabs and C# across a portfolio no single engineer owned
- **Kept the portfolio compliant and in the stores** through a platform requirement change, upgrading SDKs fleet-wide

Project Lead & Product Designer, Penn Spark *Jan 2025 – Present · Philadelphia, PA*

- **Shipped client-facing 0-to-1 products** leading a cross-functional team of **15+** designers and developers
- **Made design decisions once instead of per-person**, defining the end-to-end flows and reusable interface systems the team built inside
- **Delivered on every semester-long client cycle**, brief to user-tested release

Military Interpreter, Capital Defense Command (ROK Army) *Dec 2022 – June 2024 · Seoul, South Korea*

- **Held coordination together across government and defense organizations** as real-time English–Korean interpreter in high-security, multi-agency settings
- **Carried exact meaning where an approximation would have changed what a partner believed it agreed to**, live and in sensitive briefings

Database Engineer Intern, IT-Farm *June 2022 – Aug 2022 · Seongnam, South Korea*

- **Made semiconductor data from 20+ client companies queryable** — including Samsung, SK and LG — by designing a SQL database prototype managing **1M+ entities** in Oracle and SQLite
- **Cut retrieval time across large-scale datasets** by restructuring and optimizing the queries the team relied on

Projects

Capsule | *React, Three.js (R3F), Blender* *Jan 2025 – Apr 2025*

- **Made memories something you walk through rather than scroll** — the **3D gallery** in React Three Fiber for a collaborative time-capsule app (team of 4), with direct-manipulation placement
- **Owned how the product felt to use** across the create → contribute → unlock flow

Mini Minecraft | *C++, OpenGL, GLSL* *Apr 2026*

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

Mini Maya | *C++, OpenGL, Qt* *Feb 2026 – Mar 2026*

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops (split, extrusion, triangulation) in a Qt editor

Technical Skills

AI & Agents: Claude Agent SDK, MCP servers, Claude Code, Codex, LLM APIs (Anthropic/Gemini), agent orchestration
Languages: Python, TypeScript, JavaScript, Java, Kotlin, C++, C#, SQL
Frameworks & Infrastructure: React, Next.js, Node.js, React Native (Expo), PostgreSQL/Supabase, Docker, GCP, AWS, MongoDB